

Aryaroop Majumder

AI Engineer

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SUMMARY

AI Engineer and GenAI-focused backend engineer with 3+ years of consulting/industry experience and 4+ years of Python across professional, academic, and open-source AI/data projects. Experienced designing AI-powered applications and GenAI architectures that integrate LLMs, RAG, embeddings, vector databases, APIs, AI pipelines, and agent-based components for enterprise use cases. Hands-on with Python, FastAPI, Flask, REST APIs, SQL, PySpark, Docker, CI/CD, GCP, AWS, Azure, LLMs, prompt engineering, MCP, observability-minded health checks, and production-oriented delivery. Comfortable supporting client teams through PoCs, scoping, technical solutioning, architecture discussions, best-practice guidance, and quality/risk-focused implementation.

CORE SKILLS

Backend, APIs and Microservices: Python, FastAPI, Flask API, REST APIs, asyncio concepts, SQL, Pandas, PySpark, API design, backend data processing, microservice-oriented patterns

GenAI Frameworks and LLM Applications: LLMs, RAG, prompt engineering, embeddings, semantic search, hybrid retrieval, BM25, reranking, NLP, Natural Language to SQL, OCR, LangChain/Llamaindex familiarity, Hugging Face, OpenAI API, Anthropic API

Agentic AI, MCP and Vector Databases: agent-based components, Model Context Protocol (MCP), Claude Skills, Claude Code subagents, CrewAI working knowledge, Qdrant vector database, PostgreSQL foundations, vector indexes, data modeling for AI workloads

Cloud, MLOps and Industrial Readiness: GCP, Google AI services, AWS Lambda, AWS Glue, EC2, S3, Azure Blob, Azure VM, Docker, Airflow, CI/CD, ETL, MLOps pipeline exposure, health checks, release support, quality/security/compliance and risk awareness

Enterprise Delivery: PoCs, technical scoping, architecture reviews, audits, best-practice guidance, code reviews, mentoring, stakeholder demos, structured communication, service mindset

PROFESSIONAL EXPERIENCE

Data Science Specialist | Brainlabs | Bengaluru, India | April 2025 - Present

- Designed and delivered SearchIQ and an AI SEO product suite on GCP, combining GenAI utilities, RAG chatbot functionality, SEO analytics, reusable data-processing workflows, and CI/CD support for internal products, client-facing demos, and pitch-ready prototypes.
- Built GenAI product-tagging automation for a Fortune 500 global consumer products client by combining image signals, onsite content scraping, prompt engineering, classification logic, and validation workflows to reduce manual categorization effort and create a scalable content intelligence workflow for client teams.
- Developed a client-agnostic topical classification module for an enterprise SEO analytics initiative, categorizing search phrases across heterogeneous client sources after rapid solution evaluation, technical grooming, architecture option analysis, and stakeholder review.
- Delivered AI-enabled analytics utilities including an embedding analyzer with topic clustering, SERP feature analyzer, keyword volume analyzer, cross-linkage matching tool, and Screaming Frog OSS alternative to accelerate SEO diagnostics and content opportunity discovery.
- Prepared business one-pagers, executive decks, demo flows, and SearchIQ narratives for senior stakeholders, strategic partner pitch opportunities, Fortune 500 consumer brand pursuits, marketing mix modeling PoCs, GenAI solutioning conversations, and best-practice guidance discussions.
- Led Docker and AI knowledge-sharing sessions, supported agentic AI and PyMC MMM demos, reviewed technical approaches with data engineering teams, and helped convert ambiguous prototypes into repeatable workflows with clearer delivery ownership, delivery rigor, and quality expectations.

Associate ML Engineer | Prescience Decision Solutions | Bengaluru, India | August 2024 - March 2025

- Developed scalable PySpark ETL and data transformation workflows for Amazon using Python, Pandas, and PySpark to support backend data preparation, schema normalization, reusable transformations, and downstream AI/data workload readiness.
- Implemented advanced data quality solutions with rule-based checks, data validation workflows, exception handling, audit-friendly reporting logic, and quality controls, resolving 99.99% of identified quality issues across supported processing workflows.
- Engineered Amazon-focused big data production pipelines and optimized AWS Glue DPU usage, improving processing performance by 35% and reducing operational costs by 50% while maintaining stable delivery for data-heavy workloads.
- Deployed and supported data workflows across AWS Lambda, AWS Glue, Amazon EC2, Amazon S3, Azure Blob Storage, and Azure VM environments, partnering with engineering teams on release support, debugging, environment readiness, and production stability.

Data Science Intern | Prescience Decision Solutions | Bengaluru, India | June 2023 - July 2024

- Built a Flask API-based data quality product deployed across AWS and Azure environments, then optimized backend logic, validation rules, and service performance to improve accuracy and runtime efficiency by 30%.
- Developed an Airflow-based ML/data pipeline using open-source orchestration for scheduled ingestion, transformation, and model/data workflow execution, reducing recurring platform costs and increasing processing speed by 25%.
- Built a Natural Language to SQL chatbot with Llama 3.1, Google Flan-T5, StarCoder, Streamlit, Flask API, and Docker, enabling non-SQL users to query structured data through conversational interfaces and prototype database access workflows.

SELECTED AI AND ENGINEERING PROJECTS

- Hybrid Contextual RAG Chatbot | GitHub: Solely built a production-style RAG application with FastAPI endpoints for ingestion and Q&A, Qdrant vector search, dense retrieval, BM25 sparse retrieval, reciprocal rank fusion, LLM reranking, neighboring chunk expansion, conversation-aware retrieval, grounded answers with citations, OCR, Docker deployment, health checks, and local/remote vector DB support.

- Agentic AI / Assistant Tooling | GitHub / PyPI: Built and published yobitsugi, a Python-based security scanning and LLM-assisted remediation workflow supporting Claude Code, Codex, Cursor, Gemini CLI, Aider, OpenAI, Anthropic, Google, Ollama, configurable providers, deterministic scan/fix/validate stages, rollback logs, governance-friendly traceability, and assistant skill installation.
- Dynamic Graphing System | GitHub: Built a self-contained Dockerized Streamlit BI dashboard with Python service structure for interactive analytics workflows, giving users a lightweight way to explore business data and generate visual summaries without a heavy BI platform.
- Public Portfolio and Repository Body: Maintains a public scientific ML/data science portfolio and GitHub profile with 32 public repositories spanning RAG, Python, Docker, Streamlit dashboards, PySpark notebooks, LLM chatbots, MCP/Claude tooling, and applied ML projects.

EDUCATION

Lovely Professional University | B.Tech, Computer Science Engineering, Data Science, Machine Learning and AI | 2020 - 2024 | Phagwara, India

CERTIFICATIONS AND AWARDS

- Introduction to Model Context Protocol (Anthropic, Jun 2026)
- Certificate of Completion: Claude 101 (Anthropic, Jun 2026)
- Generative AI: Introduction and Applications (IBM, Jun 2026)
- Introduction to Artificial Intelligence (AI) (IBM, Jun 2026)
- Building Generative AI Solutions with Best Practices and Design Choices - Technical, Level 300 Advanced ML (AWS, Mar 2024)
- Foundations of Prompt Engineering (AWS, Apr 2024)
- Amazon ML Summer School 2022, Top 200 across India